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COMP4139

Machine Learning (MLE)

Exam Revision

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Overview

Date & Time	Content	Coursework
02-Oct	Introduction & Mathematics for Machine Learning	Release Assignment 1
03-Oct	Data preprocessing & feature representation	
07-Oct	Lab1 Introduction to Python	
09-Oct	Feature selection & dimensionality reduction	
10-Oct	Supervised linear models (Linear/Logistic Regression)	
14-Oct	Lab2 Data pre-processing & Assignment grouping	
16-Oct	Supervised nonlinear models(1) Kernel SVM	
17-Oct	Supervised nonlinear models(2) Decision Trees	
21-Oct	Lab3 ML models	
23-Oct	Supervised nonlinear models(3) ANN	
24-Oct	Evaluation metrics	
28-Oct	Lab4 ML models	
30-Oct	Unsupervised Learning	
31-Oct	Convolutional Neural Network (1)	
		03-Nov: Release Assignment 2
		03-Nov: Deadline submission o
04-Nov	Lab5 Evaluation	
06-Nov	Convolutional Neural Network (2)	
07-Nov	Convolutional Neural Network (Applications)	
11-Nov	Lab6 Assignment 1	
13-Nov	Practical Issues in ML	
14-Nov	Q&A Session	
18-Nov	Lab7 Tensorflow Using Colab	
20-Nov	Generative Models (Autoencoder and GAN)	
21-Nov	Reinforcement Learning (Q-Learning & Policy Gradients)	
25-Nov	Lab8 Deep Learning Models	
27-Nov	Sequence Model (RNN, LSTM)	
28-Nov	Sequence Model (Transformers)	
02-Dec	Lab9 Assignment 2	
04-Dec	Trends and Challenges in Machine Learning	
05-Dec	Exam Revision	
09-Dec	No Lab	
		12-Dec: Deadline submission o



Exam This Year

- 2 hours **ExamSys in-person invigilated** exam (not open book <https://examsys.nottingham.ac.uk/>)
- If login issues, invigilators can help you to create an account
- Date: mid-end Jan, to be determined by student service
- ExamSys has a calculator, and you can also bring your **scientific calculator**. Blank papers will be provided for calculations, so bring a pen.
- Questions types: multiple choices, multiple responses and fill-in-blank.
- Final course mark:
 - 30% Coursework (6%: assignment 1; 24%: assignment 2)
 - 70% Exam



List of Permitted Calculators

Basic Models	Scientific Calculators
Aurora HC133	Aurora AX-582
Casio HS-5D	Casio FX82 family
Deli – DL1654	Casio FX83 family
Sharp EL-233	Casio FX85 family
	Casio FX350 family
	Casio FX570 family
	Casio FX 991 family
	Sharp EL-531 family
	Texas Instruments TI-30 family
	Texas BA II+ family

Log changing base

$$\log_2 x = \frac{\log_{10} x}{\log_{10} 2}$$

Examples

$$\log_2 36 = \frac{\log_{10} 36}{\log_{10} 2} = \frac{1.556}{0.301}$$

$$\log_2 64 = \frac{\log_{10} 64}{\log_{10} 2} = \frac{1.806}{0.301}$$



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Last Year's Exam



Questions

1. How to reduce the impact of "curse of dimensionality"?



Increase the number of data samples.



Increase the number of features.



Apply feature selection methods.



Apply dimensionality reduction methods.

(3 marks)



Questions

2. Select from the below that are hyper-parameters in machine learning models.



☐ Coefficients of features in a linear regression model.



☒ Weights that balance two terms in a loss function.

☐ Weights between neurons in the Multiple-layer Perceptron Neural Networks



☒ Parameter K in the K Nearest Neighbour method.



☒ The number of hidden nodes in a Multiple-layer Perceptron Neural Network.



☒ The parameters of the Gaussian kernel used in the kernel support vector machine method.



☒ The depth of trees in the Decision Tree method.

(3 marks)



Questions

3. Which of the following statements are correct for model overfitting?



The model performs well on training set.

☐ The model does not perform well on training set.

☐ The model performs well on validation set.



The model does not perform well on validation set.

☐ The model performs well on test set.



The model does not perform well on test set.

(3 marks)



Questions

4. Which of the following describes model underfitting?



☐ Low bias and high variance.

☒ High bias and low variance.

☐ Low bias and low variance.

☐ High bias and high variance.

(3 marks)



Questions

5. If a machine learning model produces a large bias error. What of the following solutions can we use to address it?



Try a more complicated model



Try a simpler model.



Reduce model regularisation.



Increase model size if applicable.



Add model regularisation.

(3 marks)



Questions

6. What is the best visulisation method below to check if there is a class imblance issue for a categorical feature?



☐ Boxplot



☒ Histogram

☐ Scatter plot

(3 marks)



Questions

7. When shall we use one-hot encoding to represent a feature?



☐ The feature is a continuous variable and the value range is large.

☒ The feature is a categorical variable and the classes are independent.

☐ The feature is a continuous variable and the value range is small.

☐ The feature is a categorical variable and the classes are dependent on each other.

(3 marks)



Questions

8. Which of the statements are correct for the normalisation methods: Z-normalisation, Min-Max Normalisation and Vector Normalisation?



☐ Applying these methods may have negative impact to distance-based algorithms, such as KNN.



They are all linear scaling methods, so applying them will not affect the original data distribution.



Applying these methods could improve the numerical stability of the machine learning models during training.

☐ We can no longer recover the original feature values once these normalisation methods are applied.

(3 marks)



Questions

9. Which of the statements below are correct for wrapper feature selection method?



The methods require the use of labels in the training set.

☐ The methods do not require the use of labels in the training set.

☐ LASSO is a wrapper method.

☐ Chi-Square is a wrapper method.



Recursive feature elimination is a wrapper method.

(3 marks)

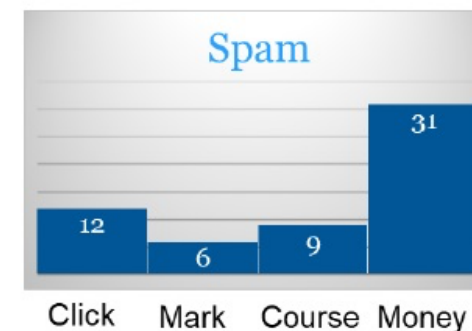
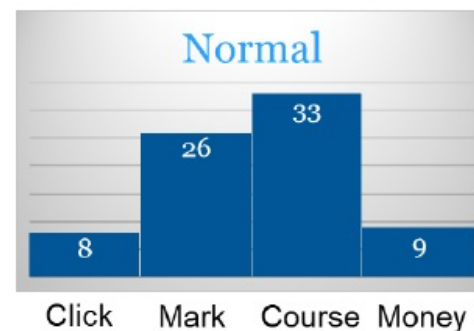


Questions

10.



$$\text{Bayes' Rule: } P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



You are asked to use Multinomial Naïve Bayes to build a model to detect spam emails using keywords.

We have collected some data and generated two discrete distributions for normal emails and spam emails based on a set of pre-defined keywords (i.e. click, mark, course, and money).

The Bayes' rule and the data are given above. We received a new email that contains "course" and "money".

Based on the provided training data using Multinomial Naïve Bayes, answer the following:

(1) What is $P(\text{course} | \text{normal})$: (round the answer to three decimal digits)

(2) What is $P(\text{money} | \text{normal})$: (round the answer to three decimal digits)

(3) What is $P(\text{course} | \text{spam})$: (round the answer to three decimal digits)

(4) What is $P(\text{money} | \text{spam})$: (round the answer to three decimal digits)

(5) What is $P(\text{normal})$: (round the answer to three decimal digits)

(6) What is $P(\text{spam})$: (round the answer to three decimal digits)

(7) Using Bayes' rule, is this email classified as Spam or Normal?

(14 marks)



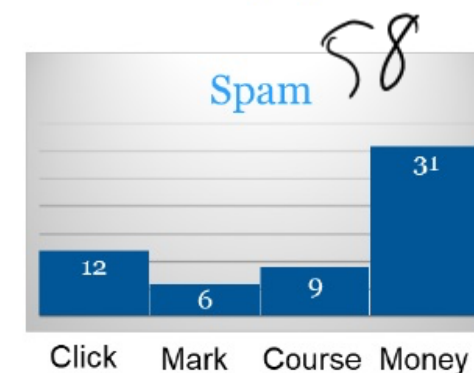
Questions

10.



Bayes' Rule: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

134



You are asked to use Multinomial Naïve Bayes to build a model to detect spam emails using keywords.

We have collected some data and generated two discrete distributions for normal emails and spam emails based on a set of pre-defined keywords (i.e. click, mark, course, and money).

The Bayes' rule and the data are given above. We received a new email that contains "course" and "money".

Based on the provided training data using Multinomial Naïve Bayes, answer the following:

- (1) What is $P(\text{course} | \text{normal})$: (round the answer to three decimal digits) $= 33/76 = 0.434$
- (2) What is $P(\text{money} | \text{normal})$: (round the answer to three decimal digits) $= 9/76 = 0.118$
- (3) What is $P(\text{course} | \text{spam})$: (round the answer to three decimal digits) $= 9/58 = 0.155$
- (4) What is $P(\text{money} | \text{spam})$: (round the answer to three decimal digits) $= 31/58 = 0.534$
- (5) What is $P(\text{normal})$: (round the answer to three decimal digits) $= 76/134 = 0.567$
- (6) What is $P(\text{spam})$: (round the answer to three decimal digits) $= 58/134 = 0.433$
- (7) Using Bayes' rule, is this email classified as Spam or Normal?

Spam

$$\begin{aligned}
 P(A|B) &\propto P(B|A)P(A) \\
 &P(N) \cdot P(\text{course}|N) \cdot P(\text{money}|N) \\
 &0.567 \times 0.434 \times 0.118 \\
 &\approx 0.03 \\
 &P(S) \cdot P(\text{course}|S) \cdot P(\text{money}|S) \\
 &0.433 \times 0.155 \times 0.534 \\
 &= 0.035
 \end{aligned}$$

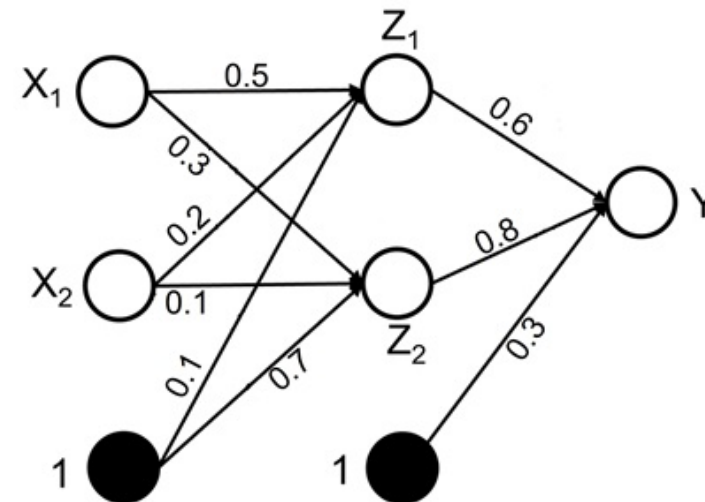


Questions

11.



Activation function at the hidden nodes Z_1 and Z_2 : $f_z(a) = a^2 - 5a$
Activation function at the output node Y is $f_y(a) = a$.



The above figure shows a ready-trained multiple-layer perceptron (MLP) neural network.

The estimated weights are noted on the edges between the nodes.

The activation functions at the hidden nodes and the output node are also provided in the above figure.

In model deployment, given the input features $X_1=2$ and $X_2=3$, calculate the output value $Y =$ (round the answer to 3 decimal digits).

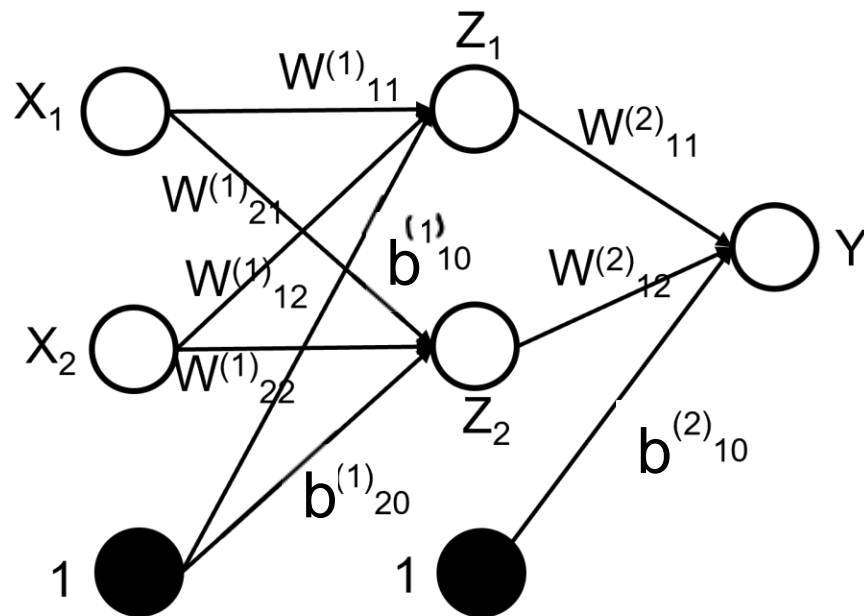
(6 marks)

Example

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$



$$(W^{(1)}_{11} = 0.10) \quad (W^{(1)}_{12} = 0.20)$$

$$(W^{(1)}_{21} = 0.30) \quad (W^{(1)}_{22} = 0.40)$$

$$(b^{(1)}_{10} = 0.05) \quad (b^{(1)}_{20} = -0.05)$$

$$(W^{(2)}_{11} = 0.25) \quad (W^{(2)}_{12} = -0.30)$$

$$(b^{(2)}_{10} = 0.10)$$

Inputs: $X_1 = 0.05$, $X_2 = 0.10$

Target: $T = 0.30$

Learning rate: $\eta = 0.1$

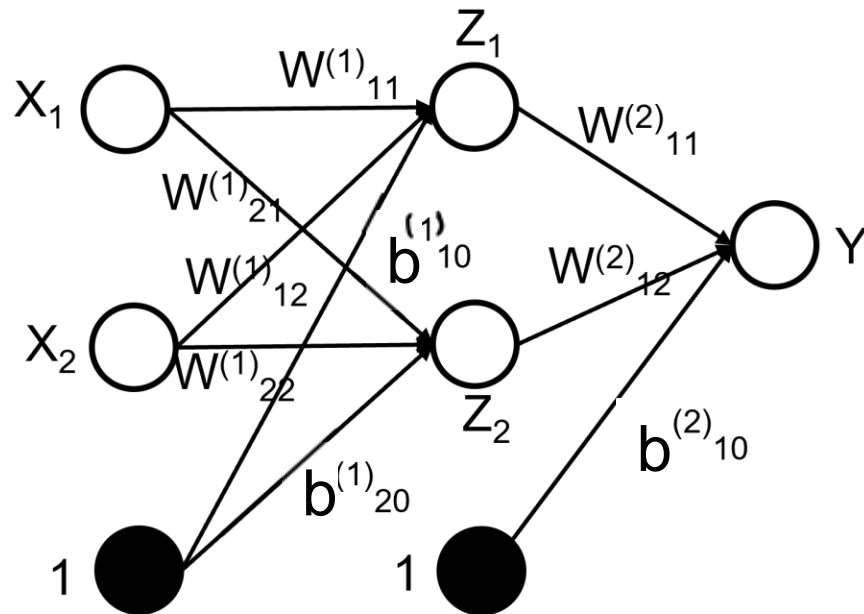
Example

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$.

Hidden Layer



$$a_{Z_1} = X_1 W_{11}^{(1)} + X_2 W_{21}^{(1)} + b_{10}^{(1)}$$

$$a_{Z_1} = 0.05(0.10) + 0.10(0.30) + 0.05 = 0.085$$

$$Z_1 = (a_{Z_1})^2 + 3a_{Z_1} = 0.007225 + 0.255 = 0.2622$$

$$a_{Z_2} = X_1 W_{12}^{(1)} + X_2 W_{22}^{(1)} + b_{20}^{(1)}$$

$$a_{Z_2} = 0.05(0.20) + 0.10(0.40) - 0.05 = 0.03$$

$$Z_2 = (a_{Z_2})^2 + 3a_{Z_2} = 0.0009 + 0.09 = 0.0909$$

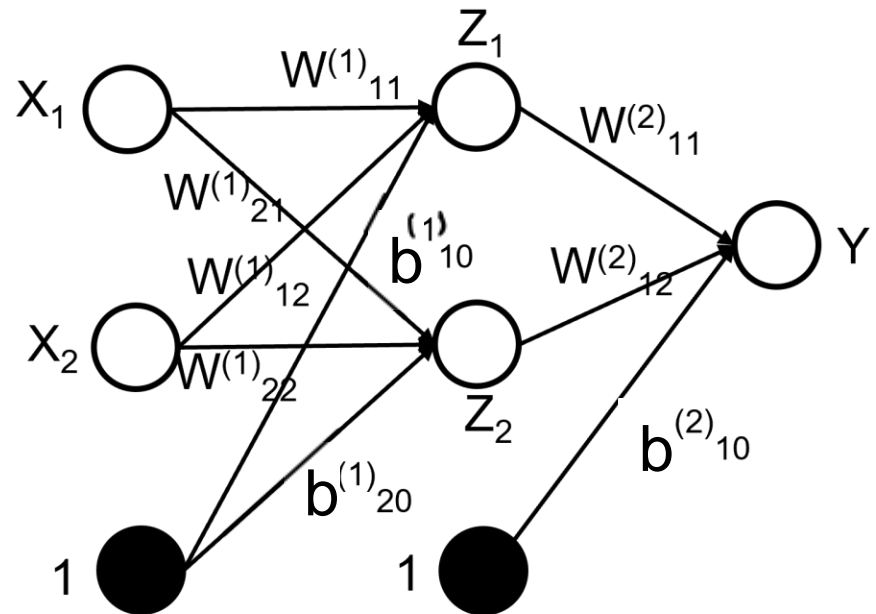
Loss function MSE:
$$L = \frac{1}{2N} \sum_{n=1}^N (Y_n - \hat{Y}_n)^2$$

Example

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$.



Output Layer

$$a_Y = Z_1 W^{(2)}_{11} + Z_2 W^{(2)}_{12} + b^{(2)}_{10}$$

$$a_Y = (0.2622)(0.25) + (0.0909)(-0.30) + 0.10 = 0.06555 - 0.02727 + 0.10 = 0.1383$$

$$Y = f_y(a_Y) = a_Y = 0.1383$$

Loss

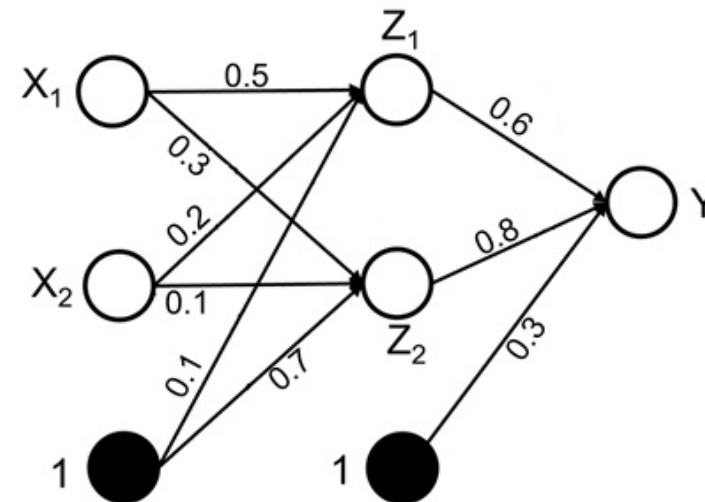
$$E = \frac{1}{2}(Y - T)^2 = \frac{1}{2}(0.1383 - 0.30)^2 = 0.01305$$

Questions

11.



Activation function at the hidden nodes Z_1 and Z_2 : $f_z(a) = a^2 - 5a$
Activation function at the output node Y is $f_y(a) = a$.



The above figure shows a ready-trained multiple-layer perceptron (MLP) neural network.

The estimated weights are noted on the edges between the nodes.

The activation functions at the hidden nodes and the output node are also provided in the above figure.

In model deployment, given the input features $X_1=2$ and $X_2=3$, calculate the output value $Y =$ (round the answer to 3 decimal digits).

(6 marks)



Questions

$$\min_{w,b,\xi_n} \frac{1}{2} \|w\|^2 + C \sum_{n=1}^N \xi_n \quad s.t. \quad t_n (w^T x_n + b) + \xi_n \geq 1$$

The loss function of soft support vector machine (SVM) method is shown above.

C in the loss function is a weighting term. How does the value of C impact the decision margin D of SVM?



A higher C results in a smaller D.

☐ A higher C results in a larger D.

(3 marks)



Questions

13. You are asked to perform cluster analysis on a given dataset.



You do not want to assume the number of clusters before the analysis.

There might be some outliers in the data. Which of the methods below is the best method to try?



DBSCAN

☐ K-means

☐ Agglomerative Hierarchical Clustering

☐ Expectation Maximisation

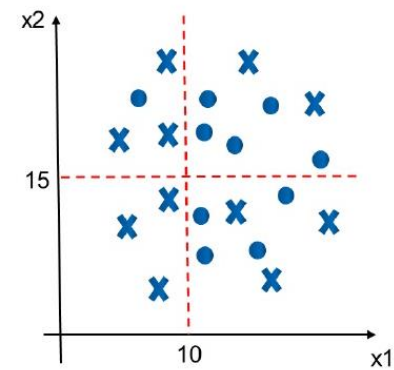
(3 marks)



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

Information Gain: $IG(S, x) = H(S) - \sum_{t \in T} P(t)H(t)$, where $S = \cup_{t \in T} t$



A dataset S was collected and presented in the figure above. Each data sample has two features x_1 and x_2 and is classified as either cross or circle.

A Binary Decision Tree needs to be built to achieve binary classification. Entropy and information gain provided above are used as the attribute selection measures.

(1) Calculate the entropy of the given dataset: (round the answer to 3 decimal digits)

(2) Calculate the entropy of the left and right sections, if $x_1 \leq 10$ is used as the decision to split the data.

The entropy of the left section is (round the answer to 3 decimal digits).

The entropy of the right section is (round the answer to 3 decimal digits).

(3) Calculate the information gain, if $x_1 \leq 10$ is used as the decision to split the data.

The information gain is (round the answer to 2 decimal digits).

(4) Calculate the entropy of the top and bottom sections, if $x_2 \leq 15$ is used as the decision to split the data.

The entropy of the top section is (round the answer to 3 decimal digits).

The entropy of the bottom section is (round the answer to 3 decimal digits).

(5) Calculate the information gain, if $x_2 \leq 15$ is used as the decision to split the data.

The information gain is (round the answer to 2 decimal digits).

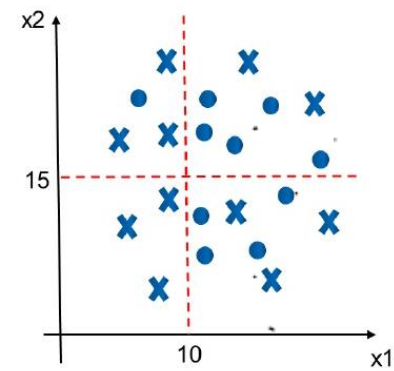
(14 marks)



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

Information Gain: $IG(S, x) = H(S) - \sum_{t \in T} P(t)H(t)$, where $S = \cup t_{t \in T}$



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A Binary Decision Tree needs to be built to achieve binary classification. Entropy and information gain provided above are used as the attribute selection measures.

(1) Calculate the entropy of the given dataset: (round the answer to 3 decimal digits)

$$H(S) = - (0.524 \cdot \log_2(0.524) + 0.476 \cdot \log_2(0.476)) = 0.998$$

$$\bullet = 10$$

$$x = 11$$

$$P_{\bullet} = \frac{10}{21} = 0.476$$

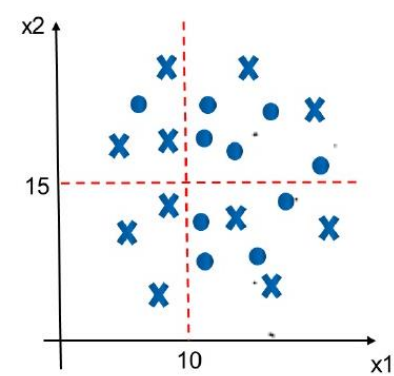
$$P_x = \frac{11}{21} = 0.524$$



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

Information Gain: $IG(S, x) = H(S) - \sum_{t \in T} P(t)H(t)$, where $S = \cup_{t \in T} t$



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A Binary Decision Tree needs to be built to achieve binary classification. Entropy and information gain provided above are used as the attribute selection measures.

- (1) Calculate the entropy of the given dataset: (round the answer to 3 decimal digits)
- (2) Calculate the entropy of the left and right sections, if $x_1 \leq 10$ is used as the decision to split the data.

The entropy of the left section is (round the answer to 3 decimal digits).

$$P_x = \frac{6}{7} = 0.857 \quad P_o = \frac{1}{7} = 0.143$$

$$H_{left} = -(0.857 \log_2(0.857) + 0.143 \log_2(0.143)) = 0.592$$

$$\bullet = 10$$

$$\times = 11$$

$$P_o = \frac{10}{21} = 0.476$$

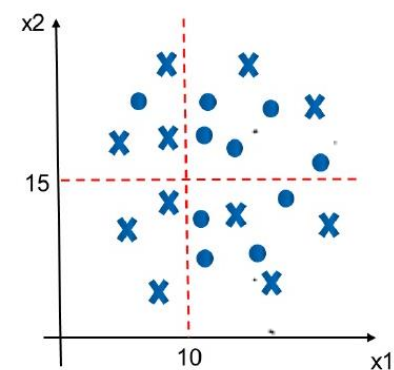
$$P_x = \frac{11}{21} = 0.524$$



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

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(2) Calculate the entropy of the left and right sections, if $x_1 \leq 10$ is used as the decision to split the data.

The entropy of the left section is (round the answer to 3 decimal digits).

The entropy of the right section is (round the answer to 3 decimal digits).

$$P_x = \frac{5}{16} = 0.3125 \quad P_{\bullet} = \frac{9}{16} = 0.5625$$

$$H_{\text{right}} = -\left(0.3125 \log_2(0.3125) + 0.5625 \log_2(0.5625)\right) = 0.910$$

$$\bullet = 10$$

$$\times = 11$$

$$P_{\bullet} = \frac{10}{21} = 0.476$$

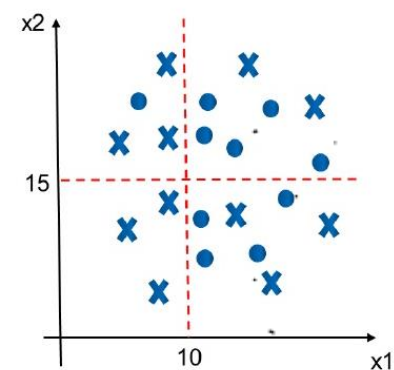
$$P_{\times} = \frac{11}{21} = 0.524$$



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

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The entropy of the left section is (round the answer to 3 decimal digits).

The entropy of the right section is (round the answer to 3 decimal digits).

(3) Calculate the information gain, if $x_1 \leq 10$ is used as the decision to split the data.

The information gain is (round the answer to 2 decimal digits).

$$\bullet = 10$$

$$\times = 11$$

$$P_{\bullet} = \frac{10}{21} = 0.476$$

$$P_{\times} = \frac{11}{21} = 0.524$$

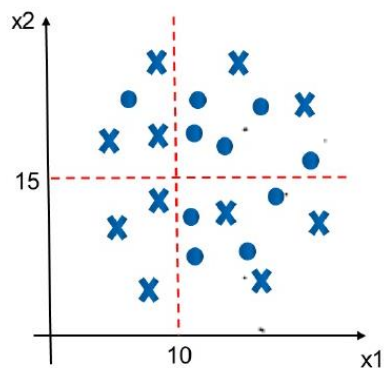
$$IG = 0.998 - \left(\frac{7}{21} \cdot 0.592 + \frac{14}{21} \cdot 0.940 \right) = 0.17$$



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

Information Gain: $IG(S, x) = H(S) - \sum_{t \in T} P(t)H(t)$, where $S = \cup_{t \in T} t$



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(2) Calculate the entropy of the left and right sections, if $x_1 \leq 10$ is used as the decision to split the data.

The entropy of the left section is (round the answer to 3 decimal digits).

The entropy of the right section is (round the answer to 3 decimal digits).

(3) Calculate the information gain, if $x_1 \leq 10$ is used as the decision to split the data.

The information gain is (round the answer to 2 decimal digits).

(4) Calculate the entropy of the top and bottom sections, if $x_2 \leq 15$ is used as the decision to split the data.

The entropy of the top section is (round the answer to 3 decimal digits).

$$P_{\bullet} = \frac{5}{11} = 0.455 \quad P_{\times} = \frac{6}{11} = 0.545$$

$$H_{top} = -(0.455 \cdot \log_2(0.455) + 0.545 \cdot \log_2(0.545)) = 0.994$$

$$\bullet = 10$$

$$\times = 11$$

$$P_{\bullet} = \frac{10}{21} = 0.476$$

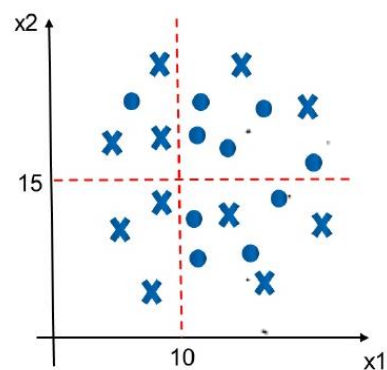
$$P_{\times} = \frac{11}{21} = 0.524$$



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

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The entropy of the left section is (round the answer to 3 decimal digits).

The entropy of the right section is (round the answer to 3 decimal digits).

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The information gain is (round the answer to 2 decimal digits).

(4) Calculate the entropy of the top and bottom sections, if $x_2 \leq 15$ is used as the decision to split the data.

The entropy of the top section is (round the answer to 3 decimal digits).

The entropy of the bottom section is (round the answer to 3 decimal digits).

$$\bullet = 10$$

$$\times = 11$$

$$P_{\bullet} = \frac{10}{21} = 0.476$$

$$P_{\times} = \frac{11}{21} = 0.524$$

$$P_{\times} = \frac{6}{10} = 0.6 \quad P_{\bullet} = \frac{4}{10} = 0.4$$

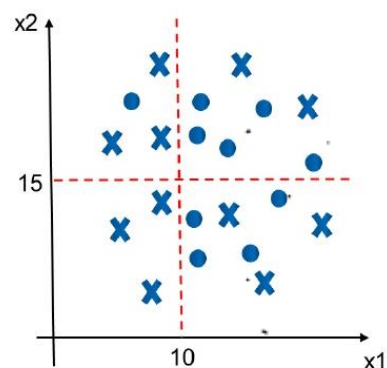
$$H_{\text{bottom}} = -(0.6 \log_2(0.6) + 0.4 \log_2(0.4)) = 0.971$$



Questions

Entropy: $H(S) = \sum_{i=1}^C -P_i \log_2(P_i)$, dataset S contains C classes.

Information Gain: $IG(S, x) = H(S) - \sum_{t \in T} P(t)H(t)$, where $S = \cup_{t \in T} t$



A dataset S was collected and presented in the figure above. Each data sample has two features x_1 and x_2 and is classified as either cross or circle.

A Binary Decision Tree needs to be built to achieve binary classification. Entropy and information gain provided above are used as the attribute selection measures.

(1) Calculate the entropy of the given dataset: (round the answer to 3 decimal digits)

(2) Calculate the entropy of the left and right sections, if $x_1 \leq 10$ is used as the decision to split the data.

The entropy of the left section is (round the answer to 3 decimal digits).

The entropy of the right section is (round the answer to 3 decimal digits).

(3) Calculate the information gain, if $x_1 \leq 10$ is used as the decision to split the data.

The information gain is (round the answer to 2 decimal digits).

(4) Calculate the entropy of the top and bottom sections, if $x_2 \leq 15$ is used as the decision to split the data.

The entropy of the top section is (round the answer to 3 decimal digits).

The entropy of the bottom section is (round the answer to 3 decimal digits).

(5) Calculate the information gain, if $x_2 \leq 15$ is used as the decision to split the data.

The information gain is (round the answer to 2 decimal digits).

(14 marks)

$$\bullet = 10$$

$$\times = 11$$

$$P_{\bullet} = \frac{10}{21} = 0.476$$

$$P_{\times} = \frac{11}{21} = 0.524$$

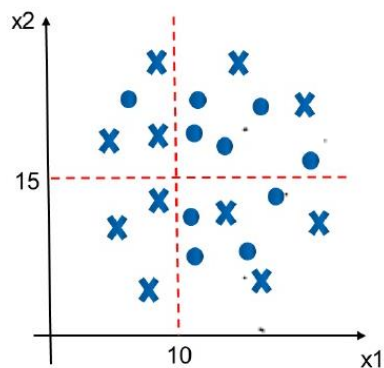
$$IG = 0.998 - \left(\frac{11}{21} \cdot 0.994 + \frac{10}{21} \cdot 0.571 \right) = 0.02$$



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(14 marks)

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$$\times = 11$$

$$P_{\bullet} = \frac{10}{21} = 0.476$$

$$P_{\times} = \frac{11}{21} = 0.524$$



Questions

15. Given an input RGB colour image with the size of 201×201 and a convolutional layer containing 64 learnable filters with the size of 5×5.



Using stride of 2 and no zero padding to calculate the convolution, what is the dimension of the output feature map?

The dimension of the output feature map is x x (width x height x channel number)

(6 marks)

$$\text{Output size} = \left\lfloor \frac{N - F + 2P}{S} \right\rfloor + 1$$

$$N = 201$$

$$F = 5$$

$$S = 2$$

$$P = 0$$

$$99 \times 99 \times 64$$



16. For deep learning model optimisation, select from the methods below that has adaptive learning rate for each parameter to be estimated. Could be more than one method.



Adam



RMSProp

☐ Gradient descent

☐ Gradient descent with momentum

(3 marks)



Questions

17. In the Transformer model, why do we need the "**Masked** Multi-Head Attention" unit rather than the "Multi-Head Attention" unit in the decoder?



Any relationship to future words need to be removed.

☐ Any relationship to past words need to be removed.

☐ Certain random words need to be removed.

☐ Non important words need to be removed.

(3 marks)



Questions

18. Select from below that is an activation function and can be used in the output layer of a neural network for a multi-class classification task.



Softmax

☐ Multi-class cross entropy

☐ Binary cross entropy

☐ Sigmoid

☐ Tanh

(3 marks)



Questions

19. Select from the statements below that best describes "mode collapse" in Generative Adversarial Network.



- ☐ The model training is extremely slow.
- ☐ During the training, the discriminator may collapse to a setting where it always predicts "real" rather than "fake".
- ☒ During the training, the generator may collapse to a setting where it always produces similar outputs with low variety.
- ☐ The model just suddenly stopped training.

(3 marks)



Questions

20. What is the key difference between Variational Autoencoder (VAE) and Generative Adversarial Network (GAN) in their learned latent spaces?



☐ The latent space in GAN is regularised by a Gaussian distribution.



☒ The latent space in VAE is regularised by a Gaussian distribution.

☐ The latent space in VAE is regularised by a uniform distribution.

☐ The latent space in GAN is regularised by a uniform distribution.

(3 marks)



Questions

	Predicted Positive	Predicted Negative
Actual Positive	80	20
Actual Negative	10	30

For model evaluation, we obtained the above table for a binary classification problem.

Based on the above table, calculate the following:

(1) True positive:

False positive:

True negative:

False negative:

(2) Classification accuracy: % (round the answer to 1 decimal place).

(3) Precision: % (round the answer to 1 decimal place).

(4) Recall: % (round the answer to 1 decimal place).

(5) Specificity: % (round the answer to 1 decimal place).

(6) Balanced classification accuracy: % (round the answer to 1 decimal place).

(9 marks)

True Positive (TP) = 80

False Negative (FN) = 20

False Positive (FP) = 10

True Negative (TN) = 30

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{80 + 30}{80 + 30 + 10 + 20} = \frac{110}{140} = 0.7857 = \mathbf{78.6\%}$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{80}{80 + 10} = \frac{80}{90} = 0.8889 = \mathbf{88.9\%}$$